

Control Systems A

Week 4 - Block Diagrams and Signal Flow Graphs

UoN - FEE 3411

Sept - Dec 2026

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What we are doing today

- **Hour 1 — block diagrams.** The three elements, the three basic connections, the canonical feedback form, and the algebra for reducing a multi-loop diagram by hand.
- **Hour 2 — signal flow graphs and Mason's gain formula.** The same information in a leaner notation, and one formula that reads the answer off the diagram without redrawing it.

Why this matters

Week 3 gave us the transfer function of a single component. This week is what happens when components are wired together — the last piece of machinery before Week 5 starts asking how these systems actually respond.

Syllabus: *Block diagrams and canonical feedback. Derivation of transfer functions from the reduction of elementary block diagrams. Construction of signal flow diagrams. Block diagram reduction using signal flow diagrams.* ELO 1, ELO 2.

From one block to many

Why we need a notation for the wiring

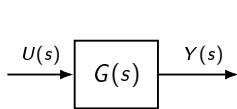
Week 3 produced transfer functions for single components: a network, a motor, a mass on a spring.

A real control system is several of them wired together:

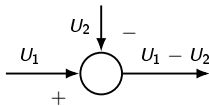
- a reference compared against a measurement,
- the difference amplified,
- the amplifier driving a motor,
- the shaft position measured by a sensor that closes the loop.

To get the transfer function of the whole thing we need a notation for the wiring, and rules for collapsing it.

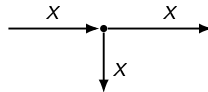
The three elements of a block diagram



(a) block: $Y = GU$



(b) summing junction



(c) pick-off point

- A **pick-off point** does not divide the signal — both branches carry X in full. The opposite of a wire junction, where current divides.
- A block diagram is a diagram of *signals*, not of power flow.

Cf. Khalil Figs. 2-31 and 2-32, pp. 40–41.

The assumption hiding in a cascade

Common pitfall — loading

Cascading G_1 and G_2 to get $G_1 G_2$ assumes connecting G_2 does not change G_1 's output — that G_2 draws no current, torque or flow from G_1 .

Two RC networks in series are the standard trap: each alone is $1/(RCs + 1)$, but wired directly together the pair is **not** $1/(RCs + 1)^2$ — the second network's input impedance loads the first.

- Cascade blocks only where a buffer amplifier separates the stages, or where the impedances make loading negligible.
- Given a *circuit* rather than a diagram: go back to Week 3's impedance method and derive the whole thing at once.

What a block diagram really is

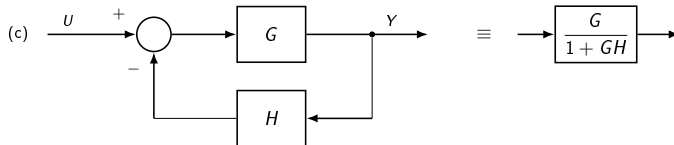
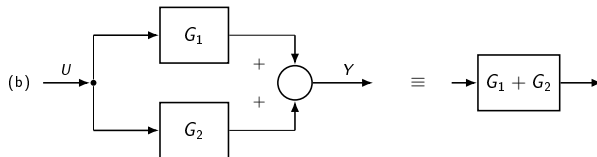
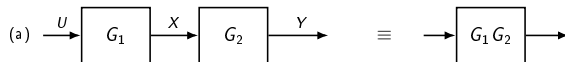
Key idea

A block diagram is a picture of a set of simultaneous algebraic equations in s — one equation per block and per summing junction. Every reduction rule this week is nothing more than eliminating an internal variable between those equations.

Consequence worth remembering under examination pressure: if a diagram ever defeats you, *write its equations out and solve them*. You will always get the right answer, just more slowly.

The three basic connections

Cascade, parallel, feedback



Cf. Khalil Fig. 2-34, p. 41; Nise §5.2, pp. 236–240.

Cascade and parallel

Cascade. $X = G_1 U$ and $Y = G_2 X$, so $Y = G_2 G_1 U$:

$$\frac{Y}{U} = G_1 G_2 \quad (\text{a product}).$$

Order does not matter algebraically — transfer functions are scalar rational functions and multiplication commutes. The physical systems certainly do not commute; swapping a real amplifier and a real motor would matter.

Parallel. Both blocks fed the same U , outputs summed:

$$Y = G_1 U \pm G_2 U \quad \implies \quad \frac{Y}{U} = G_1 \pm G_2 \quad (\text{a sum}),$$

with whatever signs the summing junction carries.

Feedback — derive it, do not quote it

With the junction *subtracting* the feedback signal, and E the signal at its output:

$$E = U - HY, \quad Y = GE.$$

Eliminating E :

$$Y = G(U - HY) \implies Y(1 + GH) = GU \implies \boxed{\frac{Y}{U} = \frac{G}{1 + GH}}$$

If the junction *adds* the feedback signal — positive feedback — the same two lines give $G/(1 - GH)$.

The sign in the denominator is the opposite of the sign at the summing junction.

The minus sign you will lose

Common pitfall

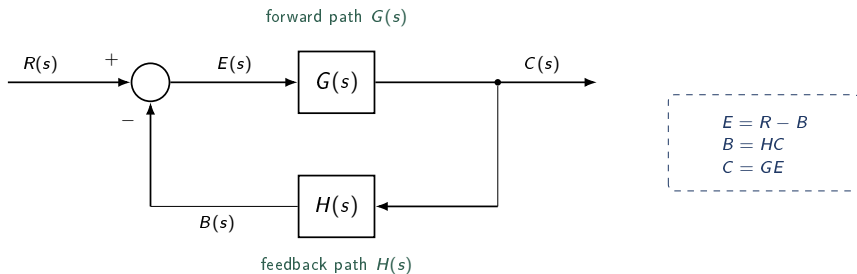
Read the sign off the *diagram*, not off memory. $G/(1 + GH)$ is for negative feedback, where the junction subtracts. A diagram with a $+$ at the feedback input needs $G/(1 - GH)$.

And that system is a completely different animal: its denominator $1 - GH$ can *vanish*, which is exactly what instability looks like.

When an answer comes out with a suspicious sign, check the summing junction first — before checking the algebra.

The canonical feedback form

The form almost every system ends up in



R reference, C controlled output, B feedback signal, E actuating error signal.

Cf. Nagrath & Gopal Fig. 2.20, p. 40.

Four transfer functions, quoted by name

$$\begin{aligned}\text{forward-path (plant):} & \quad G(s) \\ \text{open-loop (loop gain):} & \quad G(s)H(s) \\ \text{closed-loop:} & \quad T(s) = \frac{C(s)}{R(s)} = \frac{G(s)}{1 + G(s)H(s)} \\ \text{error:} & \quad \frac{E(s)}{R(s)} = \frac{1}{1 + G(s)H(s)}\end{aligned}$$

With $H(s) = 1$ the system is **unity feedback**: the measurement is taken as perfect and E becomes the true error $R - C$.

The equation the rest of the unit is about

Key idea — the characteristic equation

The closed-loop poles are the roots of

$$1 + G(s)H(s) = 0.$$

Everything downstream of this week is about those roots:

- **Week 7** — Routh–Hurwitz tests whether they are all in the left half plane.
- **Week 8** — the root locus tracks where they move as a gain varies.
- **Weeks 11–12** — the Nyquist criterion counts them from a frequency-response plot.

Open loop is GH , not G

Common pitfall

The open-loop transfer function is $G(s)H(s)$ — the gain around the whole loop. Not the forward path $G(s)$ alone, and not the closed-loop $T(s)$ with the loop cut.

So when a question says

“for the open-loop transfer function $G(s)H(s) = \frac{K}{s(s+2)}$ ”,

it is handing you the *product*, and the characteristic equation is

$$1 + \frac{K}{s(s+2)} = 0$$

directly.

Worked example 4.1 — what feedback does to the poles

Problem

Unity feedback, $G(s) = \frac{8}{s(s+6)}$, $H(s) = 1$. Find $T(s)$, its poles, its d.c. gain, and $E(s)/R(s)$.

$$T(s) = \frac{G}{1 + GH} = \frac{\frac{8}{s(s+6)}}{1 + \frac{8}{s(s+6)}} = \frac{8}{s(s+6) + 8} = \frac{8}{s^2 + 6s + 8} = \frac{8}{(s+2)(s+4)}$$

The middle step — multiplying top and bottom by $s(s+6)$ — is where the 1 in $1 + GH$ turns into the open-loop denominator. Do it carefully.

Worked example 4.1 — reading the answer

- **Open-loop poles:** $s = 0$ and $s = -6$. One at the origin, so the open-loop step response ramps away without bound.
- **Closed-loop poles:** $s = -2$ and $s = -4$ — both strictly in the left half plane. Closing the loop has made the system stable.
- **d.c. gain:** $T(0) = 8/8 = 1$; the closed-loop system follows a constant reference exactly. Week 6 calls this “a type-1 system has zero steady-state error to a step”.
- **Error transfer function:** $\frac{E}{R} = \frac{1}{1 + GH} = \frac{s(s + 6)}{s^2 + 6s + 8}$, d.c. gain 0 — the same statement, read from the other side.

Worked example 4.1 — what it means

Feedback relocates poles; it does not add them

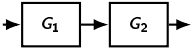
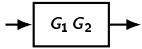
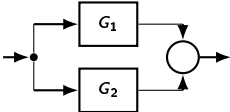
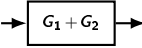
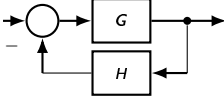
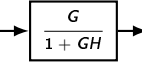
Feedback added no modes — $1 + GH = 0$ still has degree two. What it did was *relocate* the poles, $\{0, -6\} \rightarrow \{-2, -4\}$.

That relocation is the whole point of closing a loop. An unstable or sluggish plant can be made stable and quick without changing the plant itself — only the loop around it.

Choosing *where* to put the poles is what FEE3412 calls design. Finding out where they are is what this unit calls analysis.

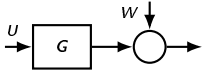
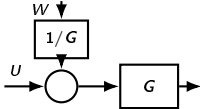
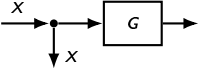
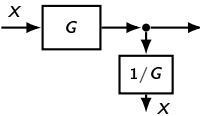
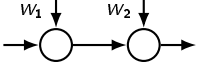
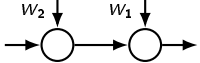
Block-diagram algebra

The rules — collapsing

Rule	Original	Equivalent
Combine blocks in cascade		
Combine blocks in parallel		
Eliminate a feedback loop		

Cf. Nagrath & Gopal Table 2.4, p. 42.

The rules — moving

Rule	Original	Equivalent
Move a summing junction back		
Move a pick-off point forward		
Interchange summing junctions		

Why the moving rules look like that

Moving a summing junction. Originally W enters *after* the block, so the output is $GU + W$. Move the junction to the block's input and whatever we add there gets multiplied by G on the way through — so add W/G :

$$G \left(U + \frac{W}{G} \right) = GU + W. \checkmark$$

Against the flow: divide. With the flow: multiply.

Moving a pick-off point. Originally the branch taps X before the block. Moved to after it, the branch would carry GX — wrong by a factor G — so the moved branch needs a $1/G$ block to undo it.

Both rules *compensate* for what the block does to the signal on the moved path.

The $1/G$ block is bookkeeping

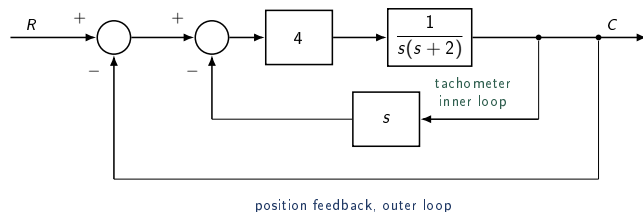
Common pitfall

A $1/G$ block introduced by one of these moves is not a physical component — and it need not even be a proper transfer function.

If $G = \frac{1}{s+2}$ then $\frac{1}{G} = s+2$: a pure differentiator plus a gain, which no hardware realises exactly.

That is fine. The $1/G$ block will cancel before the reduction finishes. It is only a problem if you stop halfway and try to read the intermediate diagram as a real system.

Worked example 4.2 — a servo with tachometer feedback



Amplifier $G_1 = 4$; motor $G_2 = \frac{1}{s(s+2)}$ (shaft angle per volt, the reduced servomotor of Week 2); tachometer measures *speed*, so $H_1(s) = s$; position measured by a unit-gain potentiometer.

Find $C(s)/R(s)$, and identify ω_n and ζ .

Worked example 4.2 — step 1, the inner loop

The inner loop is a plain negative-feedback connection of G_2 with H_1 :

$$\frac{G_2}{1 + G_2 H_1} = \frac{\frac{1}{s(s+2)}}{1 + \frac{s}{s(s+2)}} = \frac{1}{s(s+2) + s} = \frac{1}{s^2 + 3s} = \frac{1}{s(s+3)}.$$

Look at what happened

The motor's pole has moved from $s = -2$ to $s = -3$. Tachometer feedback has made the motor faster — in Week 5's language, it has added damping. That is why real servos have a tachometer, and the reason this inner loop exists at all.

Worked example 4.2 — steps 2 and 3

Step 2 — cascade the amplifier. The remaining single loop has forward path

$$G_1 \cdot \frac{1}{s(s+3)} = \frac{4}{s(s+3)}.$$

Step 3 — collapse the outer loop (unity feedback):

$$T(s) = \frac{\frac{4}{s(s+3)}}{1 + \frac{4}{s(s+3)}} = \frac{4}{s(s+3) + 4} = \frac{4}{s^2 + 3s + 4}.$$

Worked example 4.2 — reading the answer

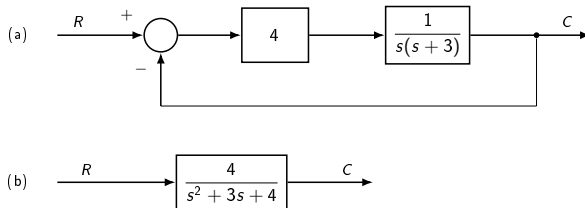
Comparing $s^2 + 3s + 4$ with the standard second-order denominator $s^2 + 2\zeta\omega_n s + \omega_n^2$:

$$\omega_n = \sqrt{4} = 2 \text{ rad s}^{-1}, \quad 2\zeta\omega_n = 3 \implies \zeta = \frac{3}{4} = 0.75.$$

The poles are at $s = -1.5 \pm j1.32$: stable, underdamped, with d.c. gain $T(0) = 4/4 = 1$.

Week 5 turns $\omega_n = 2$ and $\zeta = 0.75$ directly into an overshoot and a settling time. This week's job was only to get the transfer function they are read from.

Worked example 4.2 — the reduction in stages



(a) after the inner loop is collapsed — note the motor pole at -3 ; (b) the single closed-loop block.

When reduction gets awkward

Nested loops versus overlapping loops

Worked example 4.2 was easy because its loops were **nested**: the inner loop sat entirely inside the outer one, so collapsing it left a diagram of the same shape.

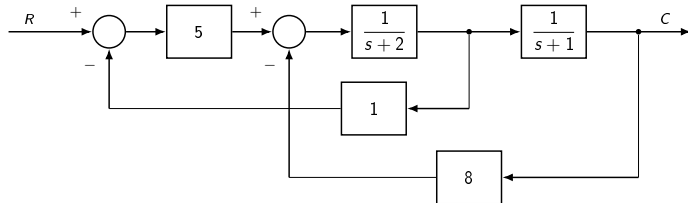
Loops can also **overlap** — each containing part, but not all, of the other.

Then there is nothing to start from

No amount of collapsing helps, because there is no *innermost* loop.

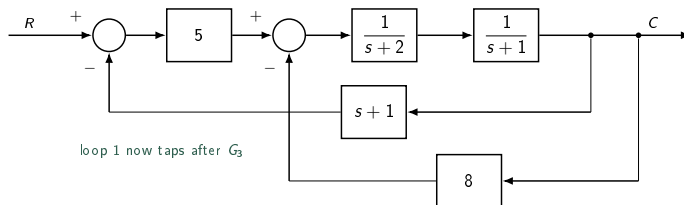
The fix is to move a junction first, which is where hand reduction starts to get error-prone — and the reason Mason's formula exists.

The smallest overlapping-loop system



Loop 1 spans G_1 and G_2 ; loop 2 spans G_2 and G_3 . Neither contains the other — they *share* G_2 .

Worked example 4.3 — move the pick-off point



Move loop 1's pick-off forward through G_3 and compensate the branch with $1/G_3 = s + 1$. Loop 1 now taps the same node as loop 2, and loop 2 sits inside it: **the loops are nested**.

Worked example 4.3 — collapse, then collapse again

Inner loop ($G_2 G_3$ with $H_2 = 8$):

$$\frac{G_2 G_3}{1 + G_2 G_3 H_2} = \frac{\frac{1}{(s+1)(s+2)}}{1 + \frac{8}{(s+1)(s+2)}} = \frac{1}{(s+1)(s+2) + 8} = \frac{1}{s^2 + 3s + 10}.$$

Outer loop, forward path $G_1 \times$ that, feedback path $H_1(s+1) = s+1$:

$$T(s) = \frac{\frac{5}{s^2 + 3s + 10}}{1 + \frac{5(s+1)}{s^2 + 3s + 10}} = \frac{5}{s^2 + 3s + 10 + 5s + 5} = \frac{5}{s^2 + 8s + 15} = \frac{5}{(s+3)(s+5)}.$$

Poles $-3, -5$, d.c. gain $1/3$ — the open-loop poles were $-1, -2$.

What that cost us

The method does not scale

One junction move, two loop collapses, one redrawn diagram — and every one of those steps is a place to lose a factor.

A diagram with *three* overlapping loops would need several moves, and each redrawing invites another error.

The answer is right. The method is the problem — and the second hour solves it.

Signal flow graphs

A leaner notation

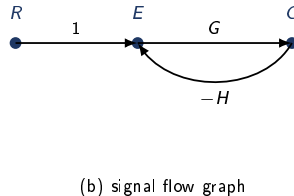
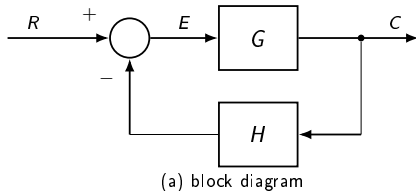
A **signal flow graph** (SFG) carries exactly the same information as a block diagram, in a notation invented by S. J. Mason so that the reduction could be replaced by a formula.

- A **node** is a signal — a dot, labelled with the signal's name.
- A **branch** is a directed line labelled with its gain. Travelling along it multiplies the signal by that gain.
- The signal at a node is the **sum** of everything arriving on branches that end there.

Why there are no summing junctions

Addition is built into the meaning of a node, and a subtraction is carried by a branch of gain $-H$ rather than a $-$ sign on a circle.

The same system, both notations



The summing junction becomes the node E , and its minus sign is absorbed into the branch gain $-H$.

Cf. Nise §5.4, p. 248; Nagrath & Gopal Fig. 2.25, p. 45.

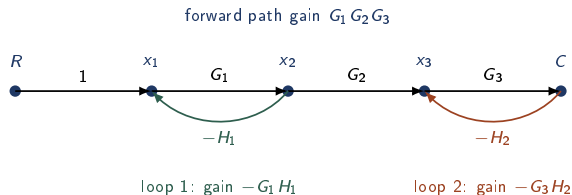
Translating between the two

Block diagram	Signal flow graph	Note
Block with transfer function G	Branch of gain G	Gain sits on the line
Signal on a connecting line	Node	Named signals become dots
Summing junction	Node (addition implied)	Signs move onto the gains
Pick-off point	Node with two or more outgoing branches	Each branch carries the full value
Input (reference)	Source node — only outgoing branches	Also called an input node
Output (controlled variable)	Sink node — only incoming branches	Add a unit branch if it also feeds a loop

The vocabulary Mason's formula needs

- A **path** — any route along branches, following the arrows, that passes through no node twice.
- A **forward path** — a path from the source node to the sink node. Its **gain** is the product of the gains along it.
- A **loop** — a closed path starting and ending at the same node, visiting no other node twice. Its **loop gain** is the product of the gains around it, *including the minus signs*.
- Two loops are **non-touching** if they share *no node at all*.

One of each



Loop 1 touches x_1, x_2 ; loop 2 touches x_3, C . The sets are disjoint, so the two loops are **non-touching** — and Mason's formula will carry a product term for the pair.

The word that costs marks

Common pitfall — “non-touching”

It means *no shared node*. Not “drawn without crossing”, and not “no shared branch”.

Two loops that meet at a single node are **touching**, and contribute no product term.

Why this one is dangerous: the error is in a term you simply *failed to write*, so nothing in the algebra flags it.

Check every pair explicitly: list each loop's nodes, then compare the lists.

Mason's gain formula

The formula

$$T(s) = \frac{\sum_k P_k \Delta_k}{\Delta}$$

- P_k — the gain of the k -th forward path, summed over *all* forward paths.
- Δ — the **determinant** of the graph.
- Δ_k — the **cofactor** of path k : Δ recomputed with every loop that *touches* path k deleted from the graph.

No redrawing, no intermediate transfer functions, one formula.

The determinant

An alternating sum over the loops — one at a time, then non-touching pairs, then non-touching triples:

$$\Delta = 1 - \sum_i L_i + \sum_{\substack{\text{pairs } i,j \\ \text{non-touching}}} L_i L_j - \sum_{\substack{\text{triples } i,j,k \\ \text{non-touching}}} L_i L_j L_k + \cdots$$

where L_i is the gain of the i -th loop, signs included.

In this unit the sum almost always stops at the pairs: triples need three loops that mutually share no node, which takes a bigger graph than any examination question here will set.

Two special cases

If every loop touches every forward path — true of most single-input single-output control diagrams — then every $\Delta_k = 1$ and

$$T(s) = \frac{\sum_k P_k}{\Delta}.$$

The canonical feedback form: one forward path $P_1 = G$, one loop $-GH$, so $\Delta = 1 + GH$ and $\Delta_1 = 1$:

$$T = \frac{G}{1 + GH} \quad \checkmark$$

— the formula of hour 1, recovered as the simplest possible case.

Which method, when

Key idea

Reduction and Mason are two routes to the same transfer function, and the choice between them is practical, not mathematical.

- **Reduction** is clearer for two or three *nested* loops, and tells a story about the physics as you go — as the tachometer pole moving from -2 to -3 did.
- **Mason** wins the moment loops overlap or forward paths multiply, because it never requires the diagram to be redrawn — and a diagram that is never redrawn cannot be redrawn wrongly.

Worked example 4.4 — the overlapping system, in one line

Same system as Worked example 4.3: $G_1 = 5$, $G_2 = \frac{1}{s+2}$, $G_3 = \frac{1}{s+1}$, $H_1 = 1$, $H_2 = 8$.

Forward paths. One: $P_1 = G_1 G_2 G_3 = \frac{5}{(s+1)(s+2)}$.

Loops. Two:

$$L_1 = -G_1 G_2 H_1 = -\frac{5}{s+2}, \quad L_2 = -G_2 G_3 H_2 = -\frac{8}{(s+1)(s+2)}.$$

Both pass through the node at G_2 's output, so they *touch*: no non-touching pair, and Δ stops after the loop-gain sum.

Worked example 4.4 — and the answer

$$\Delta = 1 - (L_1 + L_2) = 1 + \frac{5}{s+2} + \frac{8}{(s+1)(s+2)}, \quad \Delta_1 = 1$$

($\Delta_1 = 1$ because the single forward path passes through both loops.)

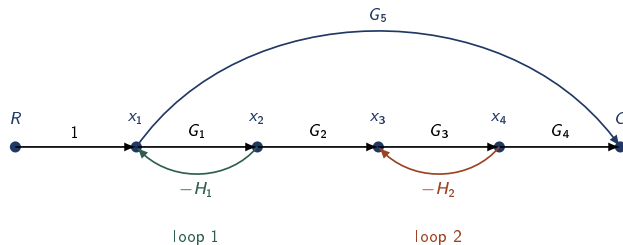
Multiplying top and bottom by $(s+1)(s+2)$:

$$T(s) = \frac{P_1 \Delta_1}{\Delta} = \frac{5}{(s+1)(s+2) + 5(s+1) + 8} = \frac{5}{s^2 + 8s + 15} = \frac{5}{(s+3)(s+5)}.$$

Compare

Exactly the answer Worked example 4.3 obtained — with no pick-off move, no redrawn diagram and no intermediate transfer functions to mislay.

Worked example 4.5 — two forward paths, one non-touching pair



$$G_1 = 1, \quad G_2 = \frac{2}{s+1}, \quad G_3 = \frac{1}{s+2}, \quad G_4 = 4, \quad G_5 = 2, \quad H_1 = 3, \quad H_2 = 2$$

Cf. Nise §5.5, p. 251.

Worked example 4.5 — paths, loops, determinant

Forward paths. Two:

$$P_1 = G_1 G_2 G_3 G_4 = \frac{8}{(s+1)(s+2)}, \quad P_2 = G_5 = 2 \quad (x_1 \rightarrow C).$$

Loops. $L_1 = -G_1 H_1 = -3$ around x_1, x_2 ; $L_2 = -G_3 H_2 = -\frac{2}{s+2}$ around x_3, x_4 . Node sets $\{x_1, x_2\}$ and $\{x_3, x_4\}$ are disjoint — so this pair *is* non-touching:

$$\Delta = 1 - (L_1 + L_2) + L_1 L_2 = 1 + 3 + \frac{2}{s+2} + \frac{6}{s+2} = 4 + \frac{8}{s+2} = \frac{4(s+4)}{s+2}.$$

Worked example 4.5 — cofactors and answer

P_1 runs through every node of both loops, so $\Delta_1 = 1$.

P_2 leaves the chain at x_1 : it touches loop 1 but *not* loop 2, so loop 2 survives in its cofactor:

$$\Delta_2 = 1 - L_2 = 1 + \frac{2}{s+2} = \frac{s+4}{s+2}.$$

Hence

$$T(s) = \frac{P_1\Delta_1 + P_2\Delta_2}{\Delta} = \frac{8 + 2(s+4)(s+1)}{4(s+4)(s+1)} = \frac{s^2 + 5s + 8}{2(s+1)(s+4)}.$$

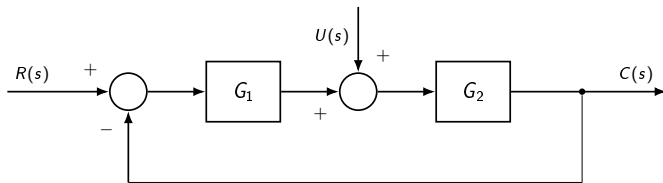
Read the numerator

Poles $-1, -4$; d.c. gain 1. The feedforward path has given the system a pair of *zeros* at $s = -2.5 \pm j1.32$, where the chain alone had none — Week 3's lesson from a new direction: a parallel route puts zeros in a transfer function, and zeros reshape the transient without moving a pole.

Systems with more than one input

Superposition

An LTI system's response to several inputs at once is the sum of its responses to each acting alone.



Method: to find the transfer function from one input, set all the other inputs to zero and reduce as usual. Then add the contributions.

One characteristic equation, several numerators

With $G_1 = 2$, $G_2 = \frac{3}{s+1}$ and unity feedback:

$$\frac{C(s)}{R(s)} = \frac{G_1 G_2}{1 + G_1 G_2} = \frac{6}{s+7}, \quad \frac{C(s)}{U(s)} = \frac{G_2}{1 + G_1 G_2} = \frac{3}{s+7},$$

$$C(s) = \frac{6}{s+7}R(s) + \frac{3}{s+7}U(s).$$

Key idea

Both share a denominator — necessarily, since both are governed by the same characteristic equation $1 + G_1 G_2 = 0$. Extra inputs change *what* the system responds to; they cannot change *how* it responds, because the poles belong to the loop, not to the input. Here U enters the loop further downstream, so it is amplified less: a unit step at U gives half the steady-state output of a unit step at R .

Where this stops in FEE3411

When the second input is something unwanted — a load torque, a supply fluctuation, sensor noise — it is called a **disturbance**.

- Computing the response, as we just did, is **analysis**: it belongs here.
- Designing a loop that *rejects* the disturbance, and handling plant uncertainty and measurement noise, is **FEE3412's** work.

Khalil §4-4 to §4-8 and Nise §7.5, §7.7 are worth reading for interest — they are not examinable in this unit.

Summary and what is next

Summary — block diagrams

- Three elements: block, summing junction, pick-off point. A pick-off point *copies* a signal; it does not divide it.
- Cascade $\rightarrow G_1 G_2$; parallel $\rightarrow G_1 \pm G_2$; negative feedback $\rightarrow G/(1 + GH)$. Cascade assumes no loading between stages.
- Canonical form: G forward path, GH open loop, $T = G/(1 + GH)$ closed loop, $1/(1 + GH)$ error. Closed-loop poles are the roots of $1 + GH = 0$.
- Feedback does not change the *number* of poles — it relocates them.
- Moving rules: a summing junction moved against the flow divides by the block; a pick-off point moved with the flow needs a compensating $1/G$.
- Nested loops reduce from the inside out; *overlapping* loops need a junction move first.

Summary — signal flow graphs and Mason

- An SFG replaces blocks with branch gains and summing junctions with nodes, absorbing the feedback signs into the gains.
- Mason: $T = \sum_k P_k \Delta_k / \Delta$, read straight off the graph.
- Δ alternates signs over loop gains, then non-touching pairs, then non-touching triples. Δ_k deletes the loops that touch path k .
- **Non-touching** means sharing *no node*.
- Several inputs: one transfer function per input, all sharing the same denominator.

The one-line version

Reduction for nested loops and physical insight; Mason for overlapping loops and multiple paths. Both give the same $T(s)$ — and Worked examples 4.3 and 4.4 proved it on the same system.

Before the tutorial

You should be able to attempt these before you walk in (notes, “Before the tutorial”):

- 1 Unity feedback with $G(s) = 12/[s(s + 8)]$: write the open-loop transfer function, the characteristic equation and the closed-loop transfer function. Are the closed-loop poles real or complex?
- 2 Which of G , GH , $G/(1 + GH)$ does “the loop gain” mean? Which sets the closed-loop poles?
- 3 A pick-off point sits before $G(s) = 4/(s + 3)$ and feeds a branch of gain 2. Redraw it with the pick-off *after* the block, and state the new branch gain.
- 4 Draw the SFG of a unity-feedback system with forward path G_1 cascaded with G_2 , then use Mason to confirm $G_1 G_2 / (1 + G_1 G_2)$.

The tutorial itself is reduction practice on multi-loop diagrams, and block diagram $\rightarrow$ SFG $\rightarrow$ Mason on the same system.

Looking ahead

- **This week finished the modelling half of FEE3411.** Week 2's differential equations, Week 3's transfer functions, this week's interconnection rules — we can now write a single $T(s)$ for a complete control system.
- **Week 5** takes the ω_n and ζ read off denominators like $s^2 + 3s + 4$ and turns them into overshoot, rise time, peak time and settling time.
- **Week 6** asks what steady-state error the response settles to; **Week 7** asks whether the roots of $1 + GH = 0$ are all in the left half plane at all.
- **Assignment 2** — issued this week, due Week 6 — covers Weeks 3 and 4: transfer functions, reduction, signal flow graphs, Mason's rule.

The boundary, stated plainly

Locating a closed-loop pole is **analysis** — FEE3411. Choosing $G(s)$ to put it somewhere better is **design** — FEE3412.

Reading for this week

Primary

Khalil, §2-3 Block Diagrams (pp. 40–49).

Secondary

Nise (6th ed.), §5.2 Block Diagrams (p. 236); §5.3 Analysis and Design of Feedback Systems (p. 245); §5.4 Signal-Flow Graphs (p. 248); §5.5 Mason's Rule (p. 251). *Khalil does not cover signal flow graphs — Nise is the primary text for hour 2.*

Further

Nagrath & Gopal (2nd ed.), §2.4 Block Diagram Algebra (p. 38), §2.5 Signal Flow Graphs (p. 45); Sundararajan, §5.1, §5.2, §5.2.1.